



Profit Sharing as a Management Strategy for a State-owned Teak Plantation at High Risk for Illegal Logging

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ABSTRACT

Illegal logging is a serious threat to plantations in the tropics. Here, we study the roles of profit-sharing in plantation management strategy in a dynamic game model. The model assumes that the owner chooses the age of the trees to be harvested and the local people choose their level of monitoring effort to prevent illegal logging with surveillance. After the trees are removed, the owner hires local people to replant young trees. We derived recursive formulas for determining the value of a site with trees of age t , and solved them by iteration. Under the pressure of illegal logging, the owner may find it profitable to share harvesting profits with the local people to enhance their surveillance effort. A higher rate of natural disturbance shortens the mean tree age, leading the owner to invest more to suppress illegal logging by increasing the profit-sharing rate. A lower future discount rate leads to older trees being harvested and decreases the profit-sharing rate. If the cost of replanting is increased, the local people are reluctant to participate in surveillance activities, and the owner increases the profit-sharing rate to encourage their participation. We discuss the policy implications of these results.

1. Introduction

Illegal logging, which is illegal activities ranging from forest ecosystem and industries, and to timber and non-timber forest products, seriously threaten forests in the tropics. In some countries such as Cambodia, Indonesia, and Bolivia, the estimated illegally logged production may exceed 80% (Food and Agriculture Organization (FAO), 2005; European Forest Institute, 2005). At both state and community levels, illegal logging occurs widely and persistently (Corbridge and Kumar, 2002; Véron et al., 2006; The World Bank Group, 2006). Plantations occupy 7% of the forest area (FAO, 2010) and play an important role in providing industrial roundwood (Jurgensen et al., 2014). Indonesia is one of the top three teak timber producers in the world, along with India and Myanmar, mainly in the state-owned forests located in Java. Indonesian teak plantations suffered a high risk of illegal logging. Teak log production decreased dramatically by widespread illegal logging, after the collapse of the Soeharto regime (1968–1998) (Awang et al., 2006; Ministry of Forestry, Indonesia, 2002, 2008).

To tackle the problem of illegal logging, a new partnership program based on sharing harvest profit with the local community was proposed

(Rosyadi et al., 2005). Profit sharing aims to suppress illegal logging by encouraging local community to join in the surveillance. The program was tested in several places and was shown to be effective in reducing damage by illegal logging (Fujiwara et al., 2012).

To understand the economic incentives underlying the community-involved forest management, we introduce a dynamic game between the landowner and the local people in the presence of an illegal logging risk. We regard the state forest company as the landowner (and the forest manager as the landowner's representative) and consider the local people, represented by an association, as a single player. We also assume that each managed teak plantation is composed of a very large number of sites and that the trees planted at each site belong to the same coeval cohort.

The purpose of the theoretical study in the current paper is to examine the idea that introducing profit-sharing mitigates the risk of illegal logging by making the local community more cooperative in the surveillance activity. We want to clarify the processes and elements that are needed for this idea to work. The model's structure was chosen to be the simplest possible one. The model was motivated by the case of Indonesian forests, in which profit-sharing seems to have successfully

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suppressed the risk of illegal logging. However, we did not intend to make the model provide a realistic description of the Indonesian forests.

A site can be cleared when (1) the forest manager decides to harvest the trees and sell the logs, (2) illegal logging takes place, or (3) a natural disturbance occurs. In deciding whether to harvest the trees, the forest manager considers the price of the timber (which increases with the age of the trees) and the risk of loss from illegal logging or a natural disturbance. Illegal logging refers to the invasion of the plantation by people or companies, who cut the trees and sell the timber illegally. The landowner hires rangers or guards to prevent illegal logging, but without the cooperation of the local people, the rangers may not be able to suppress it. The local people can help suppress illegal logging by reporting illegal logging activity when they observe it. We assume this reporting incurs no additional cost to them.

However, under some conditions, some local people may find it more profitable to allow illegal logging to take place, because after illegal logging (or a natural disturbance) occurs, the landowner needs to hire local people as workers and pay them wages to plant new trees. To reduce the labor cost, the landowner might let the workers use the land for agroforestry in lieu of wages, until the site becomes shaded by the growing trees. The community-based forest management scheme and game model is illustrated in Fig. 1.

Thus, to secure the collaboration of the local people, the state forest company may share the profit obtained by selling the harvested logs (profit-sharing). If the amount of profit-sharing is sufficient, the local people might find it more profitable to help prevent illegal logging than to overlook it. We use our dynamic game model to examine the conditions under which profit-sharing is beneficial to the owner and the optimal profit-sharing rate.

1.1. A Partnership Program for Community-based Forest Management in Indonesia

The Javanese State Forest Company (*Perum Perhutani*) has been the major producer of teak timber in Indonesia (Kollert and Cherubini, 2012). However, the teak log production by the state forest company has been dramatically decreased due to illegal logging. To deal with the illegal logging problem, the state forest company initiated a community-involved forest management scheme called PHBM (*Pengelolaan*

Sumberdaya Hutan Bersama Masyarakat) in 2001. Fujiwara et al. (2012) pointed out two major features of PHBM policies that may enhance stewardship of the forest by the local people are [1] investing to solicit joint surveillance activities by the local people to suppress illegal logging by offering them a share of the profit from the timber harvest, and [2] implementation of an agroforestry system (called *tumpang sari*) that allows the local people to cultivate cash crops in the state forests while the planted trees are too small to shade the crops.

In plantation forests in Java, agroforestry system started around 1873 and became widely spread from 1895 (Peluso, 1992; Evans and Turnbull, 2004). For the first few years after planting young trees, farmers use the plantation land for cultivating crops, such as corn, peanuts, and cassava between planting tree rows until crown closure (Fujiwara et al., 2011). The new partnership program guarantees continued use of land for agroforestry, as well as sharing profit from harvest. The local community creates an association to represent local members who join PHBN. Fujiwara et al. (2012, 2015) argued that the local people should receive benefits, such as agroforestry and profit sharing privileges, for participation in PHBM to alleviate dissatisfaction that may discourage their cooperation with other stakeholders.

2. Model

We consider a forest plantation to be composed of a large number of sites, and at each site a coeval cohort of trees has been planted. Here, the age of the trees at a site implies the number of years since the most recent clearing. We assume that the owner has numerous sites of different tree ages, which allows smooth forest management that makes profits continuously. For simplicity, we assume no heterogeneity of the sites except for tree age.

Let p_0 be the price of the trees at a site with age 1 year and $p_0 t^3$ be the price at a site with trees of age t , which represents the situation in which the value of matured trees is much higher than premature trees. This is a plausible assumption for teak forests, for which illegal logging is a serious problem. The payment c is made by the owner to workers (local people) for replanting labor just after a site is cleared. Let r be the future discount rate, and e^{-r} be the annual discounting factor that is called exponential discounter and widely accepted in economics (Samuelson, 1937; Strotz, 1955; Lucas, 1990). s is the probability that there will be no natural disturbance in a year; $(1 - s)$ is the probability a disturbance occurs in a year. L_t is the probability that illegal logging will occur at a site with trees of age t . L_t depends on the values of other parameters, as explained below.

2.1. Owner's Choice: Tree Age at Harvest

Let V_t be the value to the owner of a site with trees of age t at the beginning of a year. This value, which takes account of the current and future profits to the forest owner, is calculated with Eq. (1):

$$V_t = \max_{0 \leq \sigma_t \leq 1} \left\{ \frac{\sigma_t [(1 - \alpha)] p_0 t^3 - c + e^{-r} V_0}{+(1 - \sigma_t) [(1 - s)(1 - L_t)(-c + e^{-r} V_0)] + s(1 - L_t) e^{-r} V_{t+1}} \right\} \quad (1)$$

where σ_t is the probability that the owner chooses to cut trees with age t : $\sigma_t = 1$ indicates that the owner chooses to cut the trees, whereas $\sigma_t = 0$ indicates that the owner chooses not to cut the trees. When the owner chooses to cut the trees at a site, the logs are sold at price, which increases with tree age t . The owner may share the benefit derived from the harvest (i.e., the profit) with the local people, who receive fraction α of the profit. In addition, the owner must pay c for the labor of planting small trees at a newly cleared site. In the following year, the site value is V_0 with the trees at age 0. These terms are within square brackets and are all multiplied by σ_t .

Two terms are within the square brackets multiplied by $1 - \sigma_t$: the first term is, $1 - s(1 - L_t)$ which is the probability that the site will experience either a natural disturbance or illegal logging, multiplied by

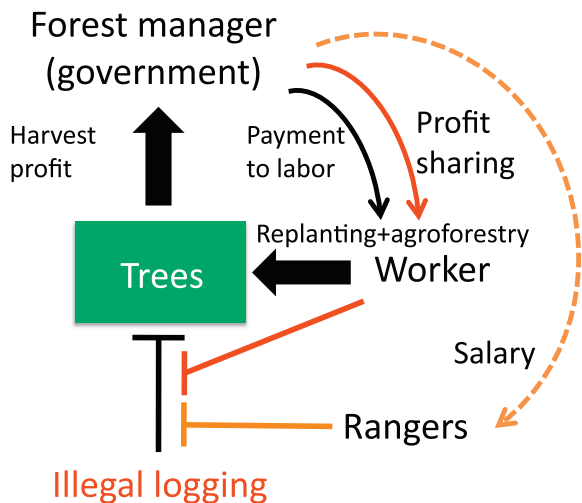


Fig. 1. The community-based forest management scheme (dynamic game model). The forest manager and the local people are the two main stakeholders. The forest manager hires rangers, who guard the forest. When a site is cleared, the manager hires local people to replant trees and perform maintenance work in the forest. In lieu of payment for their labor, the local people may be allowed to use the land for agriculture while the trees are still young (agroforestry). In addition, local people may receive some fraction of the profit when the trees are harvested in return for engaging in surveillance activities to help suppress illegal logging (profit sharing).

the initial cost of planting small trees $-c$ and the future value of the site with trees of age 0 $e^{-r}V_0$; the second term is $s(1-L_t)$, the probability that the site will not experience any disturbance, multiplied by the future value of the site in the following year when the trees will have age $t+1$. The maximum function applied to the terms within the braces implies that the owner chooses whether or not to harvest the trees to maximize her own profit.

Next, we consider the optimization of Eq. (1). Let ξ be the derivative of the terms in braces with respect to σ_t . ξ is given as follows:

$$\xi = \frac{\partial}{\partial \sigma_t} \{ \dots \} = (1-\alpha)p_0 t^3 + s(1-L_t)(c + e^{-r}(V - V_{t+1})) \quad (2)$$

Optimization of Eq. (1) is achieved by $\sigma_t = 1$, if $\xi > 0$, and by $\sigma_t = 0$, if $\xi < 0$.

Numerical analysis shows that the optimal solution always has the following form:

$$\sigma_t = 1 \quad \text{for } t \geq t^* \\ \sigma_t = 0, \quad \text{for } 0 \leq t \leq t^* - 1 \quad (3)$$

This result implies that before the trees at a site reach the optimal age for harvesting t^* , no trees should be cut. When the trees reach age t^* , all of the trees should be harvested. Hence, the system includes no trees with age greater than t^* .

The probability of illegal logging L_t is determined as follows:

$$L_t = \frac{N_{IL}}{N_{IL} + \varepsilon(h + M_t)} \quad (4)$$

where N_{IL} is the intensity of illegal logging, which is treated as a parameter in this paper, and ε is monitoring effectiveness. ε is multiplied by the total monitoring effort, which is the sum of rangers hired by the owner h and the level of monitoring effort by the local people M_t . The subscript t reflects the assumption that the monitoring effort may depend on the age of the trees at a site.

2.2. Local People's Choice: Level of Monitoring Effort

If the local people find it profitable to themselves for illegal logging to take place, then they will overlook illegal logging when it occurs. We here treat the local people as a single actor and the second player in the game. Let U_t be the profit to the local people who choose whether or not to invest in monitoring illegal logging. U_t is described by Eq. (5):

$$U_t = \sigma_t [\alpha p_0 t^3 + c + b_{agr} + e^{-r}U_0] + (1-\sigma_t)[(1-s(1-L_t))(c + b_{agr} + e^{-r}U_0) + s(1-L_t)e^{-r}U_{t+1}] \quad (5)$$

where the probability of illegal logging L_t is a decreasing function of M_t (see Eq. (4)), which the local people choose by considering their own benefit. The value of σ_t (see Eq. (3)) is not chosen by the local people but is controlled by the owner. b_{agr} is the benefit to the local people derived from agroforestry.

In the second term on the right-hand side of Eq. (5), the monitoring effort M_t should be chosen to maximize the terms in the braces. It should be large if the terms in the braces increase as M_t increases, and should be zero if the terms decrease as M_t increases. We calculate the derivative of the right-hand side of Eq. (5) with respect to M_t as follows:

$$\frac{\partial}{\partial M_t} [\text{right-hand side}] = (1-\sigma_t)s \left(-\frac{\partial L_t}{\partial M_t} \right) \varepsilon [e^{-r}(U_{t+1} - U_0) - c - b_{agr}]$$

Because L_t is a decreasing function of M_t , we have the following:

$$M_t = N_w, \quad \text{if } \eta > 0, \\ M_t = 0, \quad \text{if } \eta < 0,$$

where $\eta = e^{-r}U_{t+1} - (e^{-r}U_0 + c + b_{agr})$ and N_w is the maximum joint effort from the local people. This result, however, allows only two possibilities, no monitoring or maximum monitoring effort, which makes it difficult to calculate a numerical solution. However, people differ in their propensity to participate in surveillance activities. We represent this choice of monitoring effort by local people with the

Table 1

Model symbols are listed up (above). Four important parameters' impact to cutting timing and optima sharing rate are summarized (below).

Symbol	
V_t	The value to the owner of a site with trees of age t
σ_t	Probability that the owner chooses to cut trees with age t
α	Sharing rate of harvested tree profit with locals
p_0	The price of the trees at a site with age 1 year
c	Payment to locals for replanting tree
r	Future discounting rate
s	Survival rate from natural disturbance
L_t	Probability that illegal logging occurs at a site with trees of age t
N_{IL}	Intensity of illegal logging
ε	Monitoring efficiency
h	Rangers hired by the owner
M_t	Level of monitoring effort by locals
U_t	The profit to the locals
b_{agr}	Profit from agroforestry

Parameters	Cutting timing (σ)	Optimal sharing rate (α^*)
Higher wage (higher c)	Later	Increase
More physically disturbed (lower s)	Earlier	Increase
Higher discounting rate (higher r)	Earlier	Increase
Higher illegal logging risk (lower h)	Earlier	Increase

following graded response:

$$M_t + N_w \frac{1}{1 + \exp(-\beta\eta)} \quad (6)$$

Eq. (6) is a logit response function that is used in the application of evolutionary game theory to economics and the social sciences (McFadden, 1976; McKelvey and Palfrey, 1995; Iwasa and Lee, 2013; Lee and Iwasa, 2014; Lee et al., 2015). According to Eq. (6), as η increases gradually from a negative value to a positive value, monitoring effort increases gradually from 0 to the maximum level N_w . η reflects the difference in utility between the two cases: the growing shared profit by securing tall trees and the possible income from a cleared site. β is the sensitivity of people to the utility difference. Table 1 summarizes model parameters.

Finally, we solve the dynamic optimization problem given by Eqs. (1), (5), and (6). In this two-player game, the owner performs the optimization in Eq. (1) by choosing optimal cutting timing. Given the owner's choice of cutting age, the local people maximize the value in Eq. (5) by adopting their surveillance effort, M_t as given in Eq. (6). Thus, these three equations constitute a game theoretical model.

The solution that satisfies all three equations can be derived as follows. Let $t_{\max}$ be a number sufficiently greater than the probable age of the sites in the optimal solution. For example, let $t_{\max} = 200$. Then, let, $V_{t_{\max}} = U_{t_{\max}} = 0$ indicating that sites with age $t_{\max}$ have no value for either player. We need to specify two initial quantities, V_0 and U_0 , to calculate the optimization. We may set $V_0 = U_0 = 0$. First, we calculate the optimization of Eqs. (1), (5), and (6) for t equal to $t_{\max}^{-1}$, and then for t equal to $t_{\max}^{-1}$, and so on, repeatedly calculating (1), (5) and (6) after decrementing t by one until $t = 0$. This iteration gives new values for V_0 and U_0 , different from the initially assigned values of $V_0 = U_0 = 0$. We then replace V_0 and U_0 with these newly obtained values, and repeat the iterative procedure. The procedure is repeated, replacing V_0 and U_0 with newly obtained values, until the values U_0 and U_0 assumed in the right hand sides of Eqs. (1), (5), and (6) are closer to the values obtained in the left hand side. When they become sufficiently close, we can regard the assumed values as the correct values of V_0 and U_0 and the solution that satisfies Eqs. (1), (5), and (6).

The owner chooses the profit-sharing rate α . We assume that these are chosen to maximize the mean benefit to the owner in which the weighting of sites of different ages is equal to the stationary age distribution q_t , given as follows:

$$q_t = \frac{s^{t-1} \prod_{t'=0}^{t-1} (1 - L_{t'})}{\sum_{t'=0}^{t^*} s^{t'-1} \prod_{t''=0}^{t'-1} (1 - L_{t''})} \quad (7)$$

α should be chosen to maximize the mean future payoff $\bar{V} = \sum_{t=0}^{t^*} q_t V_t$ which is calculated using Eq. (7).

3. Results

For numerical analyses, we chose the parameters and the functional forms in the model as plausible. Because no reliable information was available, we chose the functional forms and the parameters in the model in a way that the illegal logging risk is not negligible and that introducing profit-sharing is able to suppress the illegal logging risk.

3.1. Tree Age at Harvest and Joint Surveillance Effort

Before discussing the optimal rate of profit-sharing, we explain how to calculate the optimal timing of tree harvesting chosen by the owner and the surveillance effort chosen by the local community when the profit-sharing rate is given. The optimal tree age for harvest and the level of the local people's participation in surveillance activities chosen by the owner and the local people, respectively, are found by the iterative procedure described above. The owner and the local people make decisions by assuming an agreed-upon value of the profit sharing rate used in the management scheme. Fig. 2 compares two values of V_t (Eq. (1)), the value for the owner if the forest manager chooses to harvest the trees immediately (blue line) and the value if harvesting is postponed (orange line). The optimal time to cut is the tree age where the orange and blue lines intersect. V_t increases with tree age, reflecting the fact that the price of the timber p_t increases as the trees grow because the harvested logs get bigger. The optimal tree age for harvesting is around 30 years under the assumed set of parameters with a profit-sharing rate α of 0.25. Discounting rate of 5% chosen for Fig. 2 is plausible in developing countries (Irawan and Ring, 2013; Ruslandi and Putz, 2017).

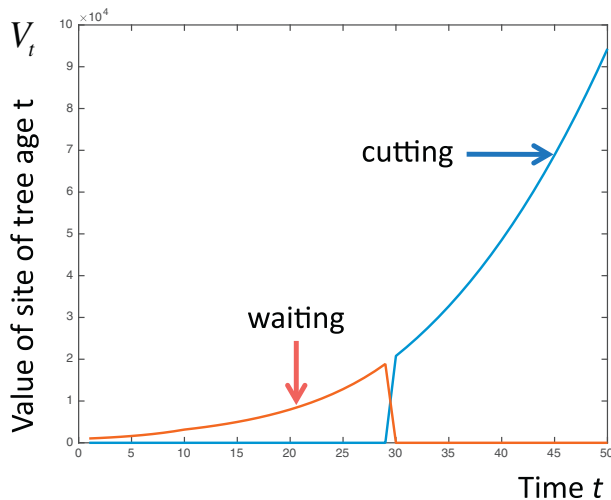


Fig. 2. Value of a site with trees of age t at the beginning of a year to the owner. The orange line indicates the value to the owner V_t when the forest manager chooses to delay harvesting the trees, and the blue line indicates the value when the decision is made to harvest the logs. The optimal tree age for harvesting is where the two lines intersect ($t = 30$). The profit-sharing rate α is assumed to be 0.25. The other parameters are $p_0 = 1$, $r = 0.05$, $e = 10$, $h = 50$, $s = 0.985$, $\alpha = 0.25$, $\beta = 1$, $c = 450$, $b_{agr} = 10^2$, and $N_W = 10^2$. The parameter values are the same in the rest of the figures unless otherwise indicated. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

When no reliable information was available for some of the parameters, we chose the values in a such a way to represent the situation in which illegal logging is not negligible compared to other risks and that introducing profit-sharing is able to suppress the illegal logging risk successfully. We call the values used for Fig. 2 as the baseline parameter set.

The local people are expected to change the level of their surveillance effort to suppress illegal logging as the trees become older (Fig. 3). The local people decide to fully cooperate when the trees reach an age of about 7 years (for example, $\alpha = 0.45$), which is younger than the optimal harvesting time chosen by the owner. Under this condition, the local people find it more profitable to secure their share of the profit from the tree harvest than to have open land for agroforestry. Thus, they adopt a high level of monitoring activity. Illegal logging intensity notably declines when joint surveillance increases because of the local people's participation. Under the assumptions of Fig. 3, the illegal logging intensity L_t is controlled at around 0.1 with only the regularly hired rangers, but it drops to around 0.03 when the local people engage in surveillance.

3.2. Profit-sharing Rate

Tree survival curve, which is based on the equilibrium age distribution q_t (Eq. (7)), for different values of α are shown in Fig. 4a. Because survival of trees at a site is affected by both natural disturbances and illegal logging, it decreases as tree age increases. The mean future profit $\bar{V}$ is the value of the site at each time step weighted by q_t . The profit-sharing rate α should be chosen to maximize $\bar{V}$.

Before choosing α , the owner needs to understand how the parameters M_t , L_t , and σ_t at each tree age are affected by α . Comparison of the effects of three different rates, 0.45, 0.25, and 0.05, on each these parameters (Fig. 3) shows that the optimal harvesting timing σ_t is not dramatically changed by the choice of σ , but the time when local people start active monitoring is greatly affected.

When α is high (0.45), the local people become engage in surveillance earlier. As a consequence, illegal logging is suppressed earlier, and the cumulative effect of the suppression of illegal logging results in increased fraction of mature trees (Fig. 4a). Fig. 4a shows the tree age distributions (survivorship) for profit-sharing rates from 0.05 to 0.35 at 0.05 increments. In general, as α increases, the fraction of mature trees increases. The mean future profit $\bar{V}$ realized from the harvest peaks at a profit-sharing rate of 0.25. Although survivorship of trees is the highest with a high profit-sharing rate, 0.45, future value $\bar{V}$ to the owner was lower than in the case with a small profit-sharing rate, $\alpha = 0.1$. In the latter case, locals will much less likely participate in monitoring (Fig. 4b).

3.3. Parameter Dependence

The increase in the price of the trees with tree age, t , favors a higher tree age at harvest, but high probabilities of natural and social (human) disturbances favor a lower tree age at harvest. In our management scheme, profit sharing is a factor that controls illegal logging, a form of social (human) disturbance. The model includes many other parameters as well, among them, the natural disturbance rate, the future discount rate, the fixed cost of replanting a vacant site, the level of hired enforcer, and the benefit of agroforestry. Here, we study how these parameters influence the optimal profit-sharing rate.

i. Rate of Natural disturbances

We examined the model for different intensity of natural disturbances. The optimal age of tree harvest was 26 years for frequent disturbances (low annual survivorship $s = 0.965$), and 30 years for an intermediate rate of disturbances ($s = 0.985$, standard value), and 35 years for infrequent disturbances ($s = 0.995$). With the same

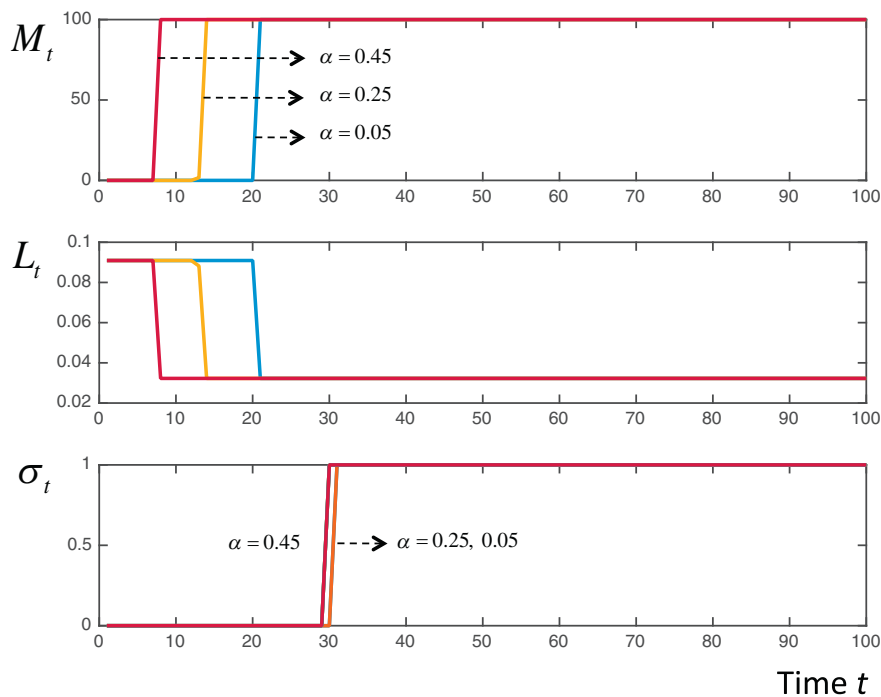


Fig. 3. The participation level of the local people in joint monitoring M_t , illegal logging pressure L_t , and harvest timing σ_t in relation to tree age for three different profit-sharing rates ($\alpha = 0.45, 0.25$, or 0.05). M_t reaches a maximum earlier, thus suppressing illegal logging earlier, when α is higher and they can expect a larger share of the profit from the harvest. The trees are also harvested sooner as α becomes larger. Illegal logging pressure is significantly reduced when the local people participate in monitoring the forest.

parameter set, when the natural is disturbance very strong ($s < 0.965$), the cutting timing becomes 1 year, indicating that risk to lose trees is too high to manage.

As the natural disturbances becomes less frequent, the optimal rate of profit sharing decreased. The optimal α was 0.25 for both $s = 0.965$

and $s = 0.985$, but the optimal α was 0.15 for $s = 0.995$.

If a natural disturbance, such as a forest fire or an insect infestation, occurs more frequently, the tree age distribution changes, this is due to the number of trees declining more quickly and the mature tree fraction becoming lower. Thus, for the owner the optimal tree age at harvest

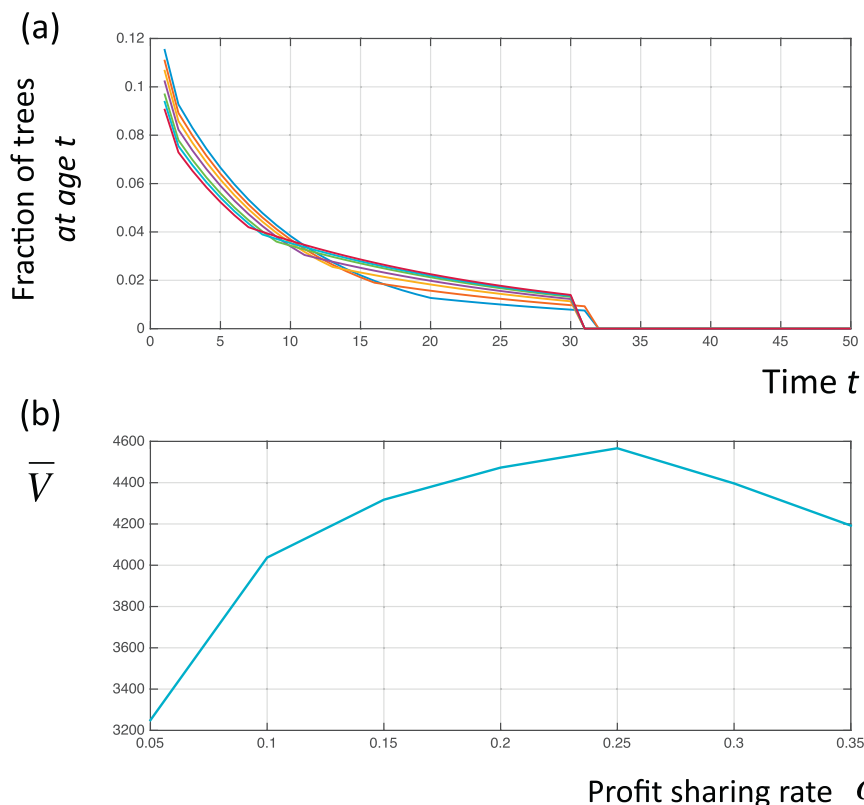


Fig. 4. Tree age distribution (survival curve) and mean future profit $\bar{V}$. (a) To determine the optimal profit-sharing rate, α is increased from 0.05 to 0.35. Smaller α creates survival curve where fractions of mature trees are low (blue colored curve applied $\alpha = 0.05$), and the mature tree fractions increase gradually as α increases. A larger profit-sharing rate increases the survivorship of matured trees more than a smaller profit-sharing rate because it increases the monitoring effort of the local people earlier. (b) The relationship between $\bar{V}$ and α shows that the optimal value of $\alpha = 0.25$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Low natural disturbance

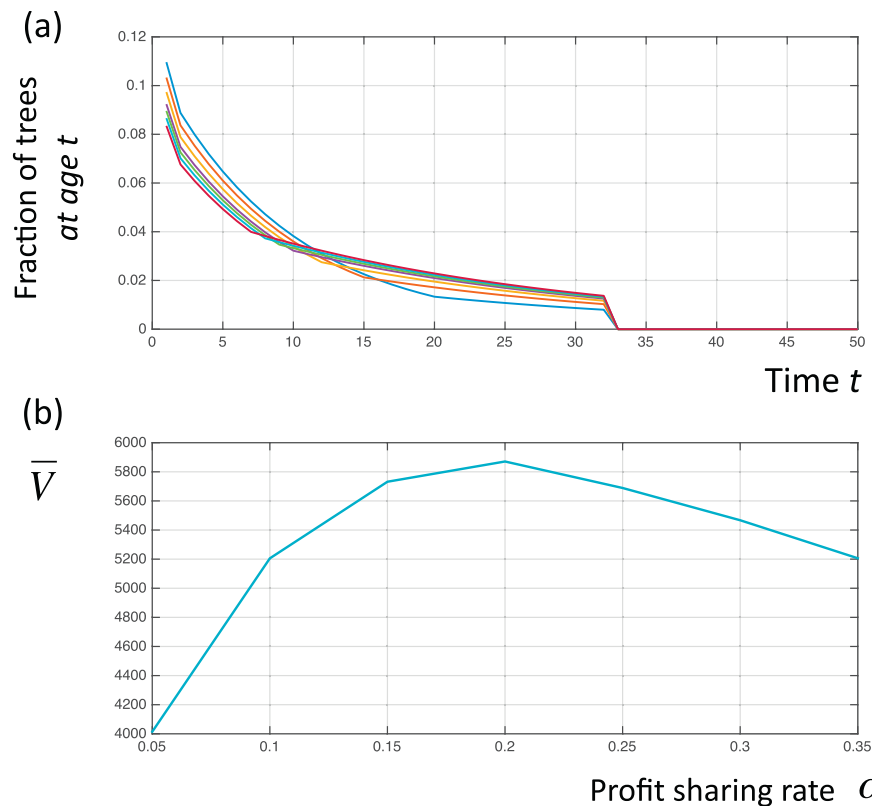


Fig. 5. The impact of a lower probability of a natural disturbance (higher s) on the optimal profit-sharing rate. (a) Tree survival curve for values of α from 0.05 to 0.35 and (b) mean future profit $\bar{V}$ in relation to α when s is high (0.99). Here, the optimal value of α to maximize $\bar{V}$ is 0.2. When the disturbance probability is higher ($s = 0.985$; Fig. 4), the optimal harvest timing is earlier (around 30 years) and the optimal value of α is increased (0.25).

becomes younger. In contrast, if natural disturbances are rare, the owner may not consider cooperation from the local people to be as important, and thus may choose a lower profit-sharing rate (Fig. 5). Even though the local people's participation begins later and the monitoring effort is lower because of the lower profit-sharing rate, the survivorship of the trees improves and the mean profit to the owner $\bar{V}$ is improved with lower natural disturbance. This result of the game indicates that, if natural disturbances are less frequent, the optimal fraction of profit-sharing decreases. This can be explained as follows: a smaller natural disturbance rate enhances the longevity of trees, enhancing the value of matured trees, which makes surveillance activity more profitable to the local people because they can receive a larger amount when trees are sold. Then, the local people become cooperative even for a smaller profit-sharing rate.

ii. Future discounting rate

Next, we examined how the future discounting rate r affects the optimal age of tree harvest. As discounting rate decreases, the optimal cutting age increased from 26 ($r = 0.08$), to 30 ($r = 0.05$, standard), and to 43 ($r = 0.02$).

We also examined how a slower rate of future discounting, a smaller r (i.e. future is more important), affects the choice of α . As r decreased, the optimal profit-sharing rate α decreased from 0.25 ($r = 0.08$) to 0.15 ($r = 0.02$). The value of α was 0.25 for r greater or equal to 0.05. If r was very large, the forest owner should quit running forests.

Because a lower discount rate also increases the future value not only to the owner but also to the local people, their willingness to become involved in any activity that enhances the future profit from the

harvest becomes stronger. Therefore, the suppression of illegal logging becomes more important to the local people than the case of a high discount rate. Moreover, it is advantageous to the owner to maintain the site longer, which means that the optimal timing of harvest becomes much later when the discount rate is lower (Fig. 6). At the same time, when the discount rate is lower, the optimal profit-sharing rate becomes smaller. This result seems counterintuitive because the owner should find it more important to prevent illegal logging. However, because the local people become more willing to engage in surveillance activity, they are willing to participate in monitoring the forest at a lower profit-sharing rate.

iii. Cost of replanting a vacant site

Finally, we examined the effect of a decrease in the fixed cost of replanting a site c on the choice of α (Fig. 7). If c is decreased by 1/9, compared with the original case (Fig. 2), the optimal harvest timing is advanced by 1–4 years and the optimal profit-sharing rate is decreased. Mean future profit $\bar{V}$ is also improved compared with that for the original value of c , and with the decreased profit-sharing rate the survivorship of mature trees are not degraded. The local people's willingness to participate in joint monitoring is influenced by both the profit-sharing rate and the replanting cost. If illegal logging takes place, they received wage income c earlier from cleared land, but they failed to receive shared profit from harvest. When this income c is fixed at a smaller value, the local people are less tempted to overlook illegal logging. Thus, a higher profit-sharing rate is not required as an incentive for monitoring participation to suppress illegal logging. Because of the lower fixed cost to the owner, combined with the smaller profit-

Smaller discount rate r

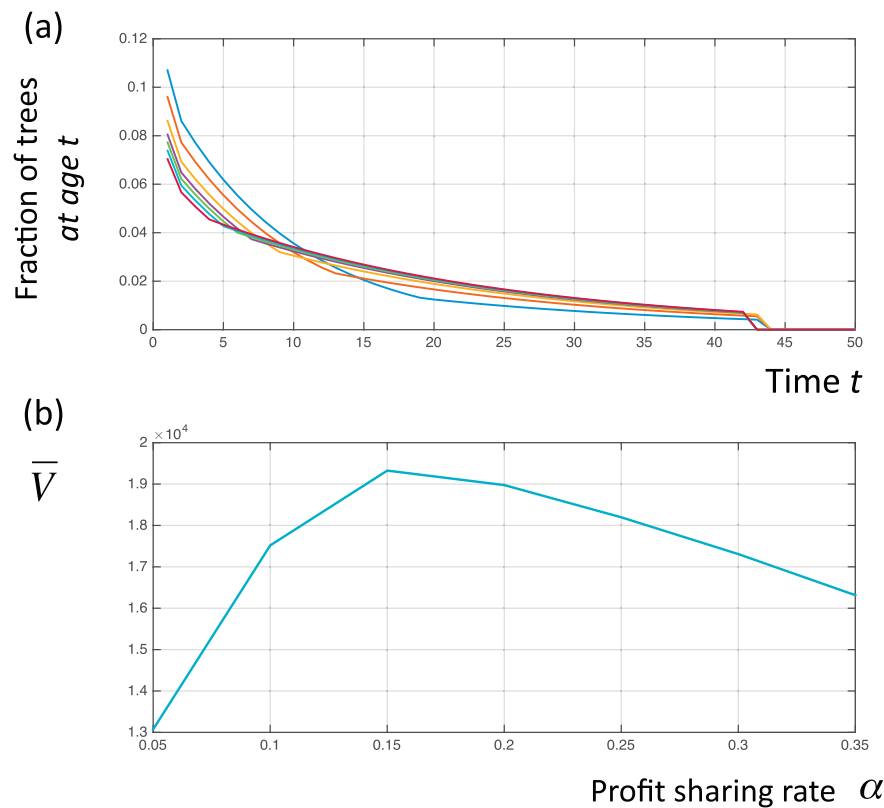


Fig. 6. The impact of a lower future discount rate r on the optimal profit-sharing rate. Tree survival curve for values of α from 0.05 to 0.35 with a smaller future discount rate r (0.02) compared with the base case (0.05; Fig. 2). The same discount rate is applied to the decisions of both the forest manager and the local people. When the stakeholders appreciate the future value more (lower r), the fraction of mature trees is increased and the optimal harvest timing is later. (b) As a result, a higher mean profit $\bar{V}$ is realized with a smaller profit-sharing rate (0.15).

Smaller replanting cost

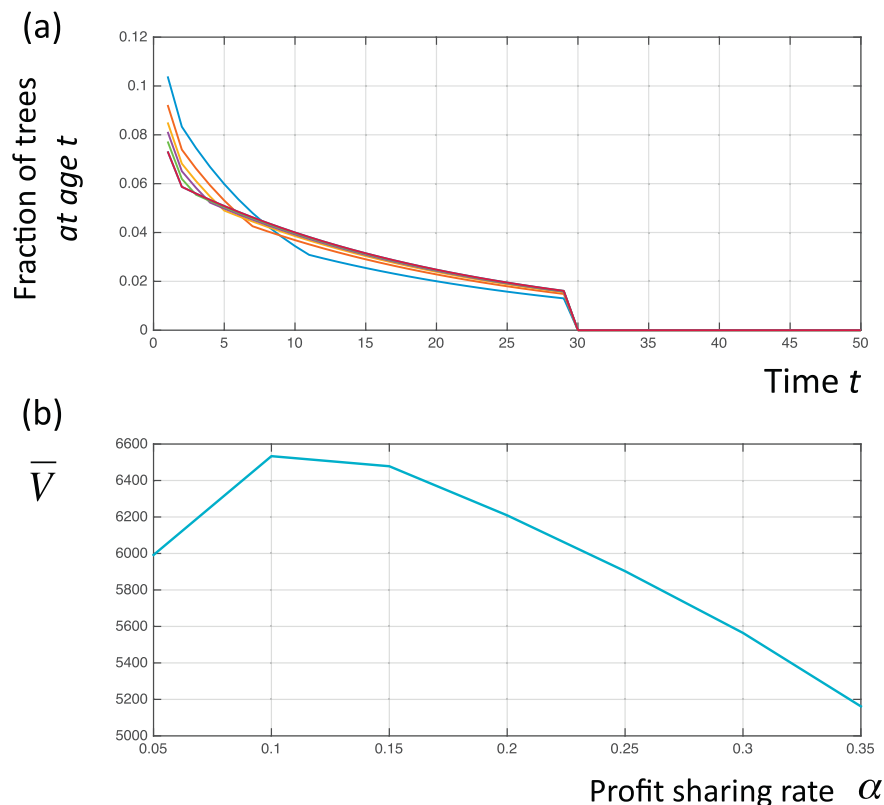


Fig. 7. The impact of a lower replanting cost c on the optimal profit-sharing rate. (a) Tree survival curve for values of α from 0.05 to 0.35 when c is reduced by $1/9$ compared with the base case (Fig. 2). With lower c , both the owner's cost and the local people's income are decreased after the trees are cut. (b) As a result, the optimal profit-sharing rate is decreased ($\alpha = 0.1$), because cleared land is less attractive to the local people.

sharing rate, the harvest timing can be advanced without decreasing the mean future profit $\bar{V}$.

As the wage c increased from 50 to 700, the optimal sharing rate α increased from 0.1 ($c = 50$) to 0.3 ($c = 700$). If wage income c becomes as large as 900, the forest owner is better off stopping plantation.

iv. The level of hired enforcers for monitoring

Forest manager may hire enforcers to suppress illegal logging. Then, the illegal logging risk is controlled by combination of enforcers' effort and locals' participation in a joint monitoring activity. If the hired effort h decreases, the locals' monitoring became more important. We test the effect of different levels of hired enforcers h to the optimal sharing rate. The optimal profit-sharing rate was $\alpha = 0.3$ when $h = 35$, it became $\alpha = 0.25$ under the standard value $h = 50$; and it became $\alpha = 0.15$ when $h = 65$.

v. The benefit from agroforestry.

Without agroforestry ($b_{agr} = 0$), the optimal cutting timing was around 30 years and the optimal profit sharing rate α was 0.2. Under the base-line parameter set ($b_{agr} = 100$), cutting timing was the same (30 years), but the optimal profit-sharing rate was larger $\alpha = 0.25$. Although we increased the level of b_{agr} by ten times as 10^3 , neither the cutting timing nor the optimal profit-sharing rate was changed. We concluded that the impact of agroforestry is smaller than other parameters.

Replanting cost c and agroforestry profit b_{agr} has an indirect effect on the cooperation of locals for surveillance in the same direction: their increase makes local people less motivated for surveillance. In addition, replanting cost has a direct effect. As shown in Fig. 7, in addition to its indirect effect to enhance cooperation smaller c (reduced fixed cost of harvesting for owner) (see Eq. (1)) may shorten the tree harvest age and reduce sharing rate. We believe this explains why change in c has a more pronounced effect to the system than change in agroforestry profit b_{agr} .

4. Discussion

Most tropical hardwood production involves deforestation and degradation of natural forests (Keogh, 2009). In Asia, timber reserves in natural forests have been severely reduced, to the point that they are unable to meet domestic needs (Odoom, 2001). The establishment of timber plantations appears to be one of the most realistic ways of conserving natural tropical forests as biodiversity hotspots while meeting increasing timber demands (Keogh, 2009). On the other hand, timber plantations change land use for people living in areas where plantations are established, and can actually lead to their participation in illegal logging or plundering of the forest to reduce their hardship. To address this situation, a community-based forest management scheme, (PHBM), has been implemented for state-owned teak plantations in Indonesia where profit sharing with local community from teak tree harvest has been initiated for illegal logging suppression.

In this paper, we have developed a mathematical model to examine the roles of profit sharing and agroforestry privileges in the management strategy for forests at high risk of illegal logging. The model is a two player game with a leader-follower structure: the forest owner first chooses the harvest age of the trees (which may be disturbed by natural phenomena or by illegal logging), and the local people, who know the chosen harvest age, then choose the level of their monitoring effort to prevent illegal logging. The model makes the following assumptions: After the trees are removed, whether by harvesting, a natural disturbance, or illegal logging, the owner hires workers from the local community to replant young trees. While the trees are young, the land can be used for crop cultivation (agroforestry), which reduces the cost for the forest manager because it compensates the local people (fully or

partially) for their replanting effort. The prospect of agroforestry, however, may discourage local people from participating in monitoring for illegal logging, which creates openings for agroforestry at plantation sites. Therefore, to create a long-term benefit from the suppression of illegal logging, the owner shares a part of the profit obtained by selling the harvested timber with the local people. Illegal logging may be prevented by hired forest guards and by local people who engage in the surveillance activities. We derived recursive formulas for determining the value of a site with trees of age t to the forest owner and to the local people, and solved them by iteration.

In the iteration process, the timing of the timber harvest, and consequently the joint monitoring effort, is adjusted (Figs. 2 and 3) while the profit-sharing rate is assumed to be fixed at a certain value. After the optimal harvest timing is decided and the joint monitoring effort in relation to tree age is determined, we can calculate the mean future benefit from the stationary tree age distribution at the plantation sites (Eq. (7)). Then, the mean future benefit for various profit-sharing rates can be compared to choose the best rate.

To understand the impacts of social and ecological parameters of our model, we investigate the effects of the discount rate of future value, the amount paid to the local people for replanting a cleared site, and the natural disturbance rate on the choice of the profit-sharing rate, while holding the values of other parameters (i.e., illegal logging pressure and the monitoring effort) fixed. We emphasize these three parameters because raising the level of payment for replanting acts to discourage involvement of the local people in illegal logging suppression. A high discount rate leads to a lower profit-sharing rate; and the natural disturbance rate is often ignored but should be measured, because the frequencies of forest fires, plant disease outbreaks, and insect infestations are not negligible.

Our model revealed that when illegal logging pressure exists, profit-sharing generally leads to higher profits for the owner, as shown by positive optimal α values. A larger profit-sharing rate increases the willingness of the local people to participate in monitoring for illegal logging, thus improving the tree age distribution (higher survivorship of mature trees) compared to the case with lower α . The three explored parameters have different impacts on the choice of the profit-sharing rate. An increased natural disturbance rate decreases the number of trees surviving to the next age step, thus reducing the mean tree age. When the natural disturbance rate is high, the forest owner should pay attention to a controllable parameter, such as illegal logging intensity, and accept an increased profit-sharing rate. An increased future value realized by a smaller future discount rate decreases the optimal profit-sharing rate because the forest owner delays the harvest and the local people engage in monitoring activities earlier when the future value of the trees is higher. The payment for replanting gives the local people an incentive to overlook illegal logging activity and makes them reluctant to participate in surveillance activities. Thus, the owner needs to increase the profit-sharing rate to solicit their cooperation if the payment for replanting increases.

There have been studies on sharecropping in agriculture, which is a form of contract between landlord and tenant under which landlord lends the land in return for the share of crop produced. Stiglitz (1974) pointed out that sharecropping allowed them to share a risk caused by large uncertainty in agriculture, for example, to unpredictability of rain fall. Eswaran and Kotwal (1985) addressed its role to attenuate moral hazard between landlord and tenants. In agriculture, productivity depends strongly on the tenant's abilities to supervise labor, and the landlord's ability of information collection and management, neither of which is easily observed. Hence, under fixed wage or rent system, moral hazard in both sides may occur in their decision on their investment of effort to agricultural production, which can be mitigated by sharecropping.

Note that the model discussed in this paper is different from sharecropping. In our model of the game of forest owner and local community, the risks including illegal logging and natural disturbance

are covered only by the owner. In addition, the model includes no element of moral hazard, because the local community is not committed to surveillance activity. Reporting observed illegal logging events is a simple voluntary cooperation.

A teak plantation must be maintained for many years before it is harvested, and the price of the timber is very high. Thus, the risk of illegal logging is very high, making profit-sharing an important policy for the forest owner. However, profit-sharing may not be important for all types of plantations. For example, acacia plantations have been established to provide raw materials to the pulp and paper industry (Arisman and Hardiyanto, 2006; Jauhainen et al., 2012). In contrast to teak, acacia trees grow quickly and their price is relatively low. Therefore, by comparison, an acacia plantation is less likely to be the target of illegal logging and profit-sharing is less important. We adopted a price-age function in order to reflect the situation in which matured trees are more highly valued than premature trees. Studies of cases in which different price-age functions (e.g. saturating, less sharply increasing with age) should be an important target of future theoretical study.

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